

YUXIA DING

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EDUCATION

University of Science and Technology of China	2023 – 2027(Expected)
• School of Management • Major: Statistics • GPA: 3.68/4.3 • Rank: 6/37	

SCHOLARSHIPS AND AWARDS

Outstanding Student Scholarship (Silver Award), USTC	2025
Outstanding Student Scholarship (Gold Award), USTC	2024
Chinese Mathematics Competitions for College Students (Provincial Second Prize)	2024

PROJECTS

Exact Search and Monte Carlo Methods for Network Likelihood Estimation

Advisor: Prof. Chenlei Leng, Department of Applied Mathematics, The Hong Kong Polytechnic University

- **Project Description:** Developed scalable computational methods for conditional likelihood estimation in directed reciprocal and undirected network formation models, addressing the computational and memory costs of exhaustive tetrad enumeration.
- **My work:**
 - **Exact Search:** Developed structure-aware algorithms that identify all informative tetrads directly from observed dyad states, reproduce exactly the full conditional likelihood, and reduce the computational-complexity order under sparse-network asymptotics.
 - **Direct Monte Carlo:** Designed a stratified sampling method that approximates the full criterion without enumerating or storing all valid tetrads, and established its conditional unbiasedness and convergence properties.
 - **Numerical Experiments:** Confirmed the losslessness and improved complexity scaling of both exact search algorithms over exhaustive enumeration, and validated accurate Direct Monte Carlo estimation on large networks.
 - **Research Output:** Completed the theoretical analysis, algorithmic implementation, and numerical validation; a manuscript based on this research is currently in preparation.

Electricity Market Dynamics under Renewable Integration (Course Project)

- **Project Description:** Analyzed 32,849 hourly observations from the DE-AT-LU day-ahead electricity market (2015–2018), focusing on load forecasting, renewable-energy interactions, price dynamics, and volatility.
- **My work:**
 - **Forecasting:** Developed an enhanced SARIMAX model incorporating renewable generation, calendar effects, and Fourier harmonics, reducing out-of-sample MAPE from 18.73% to 11.45%.
 - **Market Dynamics and Risk:** Applied VAR, Granger-causality analysis, impulse-response functions, and an ARX(7)–GARCH(1,1)-skew- t model to quantify market interactions, persistent volatility, heavy tails, and asymmetric downside risk; found that supply–demand fundamentals explained 28.8% of 20-day price uncertainty.

TECHNICAL SKILLS

Programming: Python, R, C, LaTeX **English:** TOEFL iBT 102 (R29/L21/S27/W25)

EXPERIENCE AND LEADERSHIP

Summer Research Intern, The Hong Kong Polytechnic University	Jul. 2026 – Present
Teaching Assistant, Practical Statistical Software, USTC	Spring 2026
President of Student and Alumni Charity Club, USTC	Jun. 2024 – Jun. 2025